

Manual

HYDRA Messaging Services SIS-HMS 4.0_{pe}

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1 HYDRA Messaging Services (HMS)

Overview

Purpose

HYDRA Messaging Services is an application for creating, sending and editing messages. The messages are exchanged between the MES clients.

With HYDRA Messaging Services, the users of the MES clients can communicate more efficiently. The users communicate directly with each other. This avoids unnecessary delays in the organization of processes and workflow.

Implementation notes

HYDRA Messaging Services is of use, if you have the following objectives:

- You want to increase the efficiency of the internal communication.
- You want to improve the workflow.

Integration

The application is based on the HYDRA escalation management. Creating, answering, forwarding or closing messages are functions that follow the mechanisms of the escalation management. Possible restrictions that are valid for the escalation management also apply to the use of this application.

You need a separate license, if you want to use the HYDRA Messaging Services for SMA.

Features

- You can create and send messages via the MES clients MOC and AIP2.
You can send the messages to the following entities:
 - A person
 - A function group
 - A terminal(the exchange of messages between two terminals is not intended)
- You can additionally send messages by e-mail.
- You can receive messages from the MES clients MOC, AIP2 and SMA.
- You can mark messages read.
- You can answer the messages at the MES clients MOC or AIP2.
- You can forward messages to another communication partner.
- You can mark the message history as closed via a specific function.

- You can configure function groups.

2 Messages (MOC)

Overview

Menu	Information management → Postings → Messages
Transaction code	hms
Function authorization	hms

Purpose

With the application "Messages" you can create and send messages. The application provides the following functions:

- Create new messages
- Mark messages read
- Answer messages
- Forward messages
- Close messages

Integration

The application "Messages" allows the communication with other persons. The following clients offer the possibility to send and receive messages:

- AIP2
- MOC
- SMA (requires a separate license)

The application is based on the HYDRA escalation management. Creating, answering, forwarding or closing messages are functions that follow the mechanisms of the escalation management. Possible restrictions that are valid for the escalation management also apply to the use of this application.

Requirements

The implementation of the HYDRA Messaging Services (HMS) includes the necessary programs and configurations.

You have linked HYDRA users of the [User administration](#) to persons of the [HR master](#). These users/persons can exchange messages. If you want to send messages to function groups, you have to generate the groups beforehand in the application [Function groups](#).

If you want to send a message to a terminal, the MOC does not require specific configurations.



In general, you can send a message to all persons, function groups or terminals. A specific workflow is not defined, i.e. it is not defined who can send a message to which recipient. The system does not guarantee that the recipient reads a message.

Selection criteria

In general, the function shows all active message histories i.e. that are not closed. In addition, the function shows all messages (histories) that were closed within the last <number> days:

Showing messages closed within the last <number> days

In addition to the active message histories, the function shows the messages marked closed within the last <number> days.



The system checks the time a message was closed. The time is displayed in the column *End* of the detail application *Overview*.

You can specify how long closed messages (histories) are shown. Two settings define this period: The entry of the <number> of days and the time the data is retained in the online data area of the escalation management. The document [MBL_ESK_Archiving.pdf](#) provides further information on archiving of messages in the escalation management.

Toolbar

The application offers the following functions. The chapter [Editing functions](#) includes further information on the individual functions.

The buttons are context sensitive.



New message

Function authorization: hms.new

Use this button to create a new message.



Read

Function authorization: hms.read

Use this button to mark a message read. The button is only active, if a message addressed to the user is selected in the detail application *Messages* and if the user has not yet marked the message read.



Clicking on this button does not open a dialog. The message is directly marked read.

Answer

Function authorization: hms.answer

Use this button to answer a message.

The button is only active, if a message addressed to the user is selected in the detail application *Messages*. The status column shows the symbol  and the column *Person* (category *Recipient*) shows the personnel number. It is the number that the user administration has assigned to the logged in MOC user.



Forward

Function authorization: hms.forward

Use this button to forward a message.

The button is only active, if a message addressed to the user is selected in the detail application *Messages*. The status column shows the symbol  and the column *Person* (category *Recipient*) shows the personnel number. It is the number that the user administration has assigned to the logged in MOC user.



Close

Function authorization: hms.close

Use this button to close a message history.

The button is only active, if the message history has not yet been closed.



Each person involved in the message history can close the message history. This is not restricted to the person who has created a new message.

When you close a message, it is automatically marked read. When you close a message, the time of the last message is updated. This message history is now shown on top (in case of descending chronological order).



Overview

Use this button to expand the message header in the detail application *Overview*. The expanded row shows the text of the first message of the message history.



Messages

Use this button to expand the messages in the detail application *Messages*. The expanded row shows the text of the message.

Detail application Overview

The detail application *Overview* shows all messages that were sent by or to the logged in MOC user.

If the logged in MOC user has not yet marked a message *Read* in the detail application *Overview*, the data row is highlighted in **bold**. The detail application [Messages](#) indicates which message of the whole message history has not yet been read. You can also see who has marked a message read. If the logged in user has not read a message of the message history, you can see this in the detail application *Overview*. If one of the other communication partners has not read a message of the message history, this message is only marked unread in the detail application *Messages*.

The columns available in the table are described below. Use the column selection function to add columns that are not displayed by default. The actual order of the fields described and the assignment to a category need not comply with the description below.

No title

No title

The column *No title* shows the status of the message history. Possible entries in the column *No title*:

Symbol	Meaning
	The message history is closed.
	The message history is active and still in process.

Unread messages

If the message history includes at least one unread message for the MOC user, the checkbox is checked ☒.

Status

The column *Status* shows the status of the message history. Possible entries in the column *Status*:

- Completed
 - The message history was closed.
- In process

- The message history is active and still in process.

Reference

The column *Reference* displays the internal identification of the message history in the database table *esk_event_msg*.

Final comment

Once the message history is closed, the column *Final comment* shows the comment of the message history.

Last message

Date/Time

The column *Date/Time* shows date/time of the most recent message of the message history.

Communication partner

The column *Communication partner* shows the last communication partner of the message history.

Subject

Subject

The column *Subject* shows the subject of the message that was entered on creating a new message.

Text

The column *Text* shows the text of a message that was entered on creating a new message.

Start

The column *Start* shows the Time/Date of the first message of the message history.

End

Once the message history is closed, the column *End* shows the point in time when the message history was closed.

Detail application Messages

The detail application Messages shows all messages of a selected message history.

By default, the messages are in descending chronological order (most recent message on top). If you send a message to a function group with several persons, the display order of the recipients is random.

If a message is unread, one data row is highlighted in **bold**. A message is unread, if:

- the logged in user has not yet marked the message read.

- the recipient of the message has not yet marked the message read.

The columns available in the table are described below. Use the column selection function to add columns that are not displayed by default. The actual order of the fields described and the assignment to a category need not comply with the description below.

Status

No title 1

The column *No title 1* displays symbols that provide information on the history of the communication. The symbols and their meaning are:

Symbol	Meaning
	The logged in user has received the message.
	The logged in user has sent the message.
	The message is part of a message history that was created before being forwarded to the logged in user OR the sender has sent the message to a function group including further persons than the logged in user.

No title 2

The column *No title 2* displays symbols that provide information on the status of the message. The symbols and their meaning are:

Symbol	Meaning
	The message is read.
	The message is unread.

Message

Date/Time

The column *Date/Time* shows the point in time when the corresponding message was created in the message history.

Communication partner

The column *Communication partner* shows the communication partner of the message.



If a message has been forwarded, the recipient of the forwarded message is displayed as communication partner.

Type of communication partner

The column *Type of communication partner* shows the type of the communication partner.

Sender/recipient

The column *Sender/recipient* shows an abbreviation that identifies the communication partner of a message as sender or recipient.

R – Recipient

S – Sender

Text

The column *Text* shows the text of a message.

Reference

Reference

The reference is the internal identification of a message (being part of a message history) in the database table *esk_event_msgdet*.

ESK reference

The ESK reference is the internal identification of the message history in the database table *esk_event_msg*.

Sender

Type of sender

The column *Type of sender* shows the type of the sender that has sent the message.

- P – Person
- T – Terminal

Person

If the sender of a message is a person, the column *Person* displays the personnel number of the sender.

Terminal

If the sender of a message is a terminal, the column *Terminal* displays the terminal number of the sender.

Recipient

Type of recipient

The column *Type of recipient* shows the type of the recipient of a message. The following entries are possible:

- Terminal
- Person
- Function

Person

If the recipient of a message is a person, the column *Person* displays the personnel number of the recipient.

If the recipient of a message is a function group, the message is "resolved" and sent to every person of the function group. The column *Person* then displays the personnel number of each person.

Terminal

If the recipient of a message is a terminal, the column *Terminal* displays the terminal number of the recipient.

Function

If the message is sent to a function group, the column *Function* shows the function group.

Person currently logged in

Person currently logged in

The column *Person currently logged in* shows the personnel number of the person currently logged in.

Editing functions

The following field descriptions refer to the functions that you can start via the toolbar. Depending on the corresponding function, some fields can be edited, some can be read only.

Overview of the fields you can edit in each function

	New message	Answer	Forward	Close
Type of recipient	✓	read-only	✓	not available
Recipient (Person/Function/Terminal)	✓	read-only	✓	not available
Subject	✓	read-only	not available	not available

	New message	Answer	Forward	Close
Text/Comment/Final comment	✓	✓	✓	✓
E-mail	✓	✓ / read-only for terminal	✓	not available

Field descriptions

Type of recipient

In the field *Type of recipient* you can select the type of recipient. Depending on the selection made, the field shows *Person*, *Function* or *Terminal*.



The search function of the fields *Person/Function/Terminal* does not filter. The application shows all created persons, function groups and terminals.

The system does not verify, if the recipient can display the message. Persons without sufficient authorizations or clients that do not support the messaging function might be recipients of a message without being able to display the message.

Person

In the field *Person* you enter the unique personnel number of the recipient.

Function

In the field *Function* you enter the name of the function.

Terminal

In the field *Terminal* you enter the terminal number of the receiving terminal.



Not only can the shop floor terminals AIP2 receive and send messages.

Subject

In the field *Subject* the subject of a message is entered.



You can use a maximum number of 50 characters.

Text

In the field *Text* you enter the message content.



You can use a maximum number of 600 characters.

E-mail

If you select this function, an e-mail is sent in addition to the message.



You can only use the option *E-mail* in connection with the types of recipient *Person* and *Function*. The necessary configurations for sending e-mails require the escalation management license. The configurations are described in the documentation dealing with the escalation management.

Comment (only Forward)

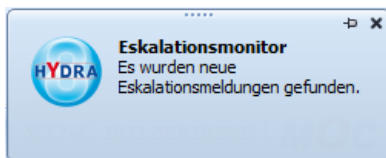
If you forward a message, you can use this field. You can enter a comment on the forwarded message in this field. The original message content is not changed.

Comment (only Close)

If you close a message history, you can use this field. You can enter a final comment in this field that is displayed in the column *Final comment* of the detail application [Overview](#). The original message content is not changed.

Further notes on processing

When the logged in MOC user receives a new message, a pop-up appears on the bottom right.



If the user clicks on the text in the pop-up, the user is led to the MOC application *Current escalations*. The messages are listed in the escalation HMS.MESSAGE.



The escalation applications do not support the function of sending messages. Only use the application *Messages* to this end.

3 Messages (AIP)

Overview

Purpose

You can create and send messages with the application "Messages" on the AIP2. The application provides the following functions:

- Create new messages
- Mark messages read
- Answer messages
- Forward messages
- Close messages

Integration

The application is based on the HYDRA escalation management. Creating, answering, forwarding or closing messages are functions that follow the mechanisms of the escalation management. Possible restrictions that are valid for the escalation management also apply to the use of this application.

Requirements

The implementation of the HYDRA Messaging Services (HMS) includes the necessary programs and configurations.

You have linked HYDRA users of the [User administration](#) to persons of the [HR master](#). These users/persons can exchange messages. If you want to send messages to function groups, you have to generate the groups beforehand in the application [Function groups](#).



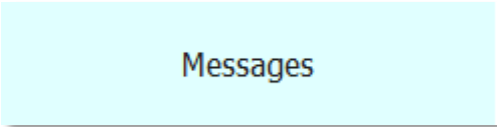
In general, you can send a message to all persons or function groups. The exchange of messages between terminals is not intended. A specific workflow is not defined, i.e. it is not defined who can send a message to which recipient.

The system does not guarantee that the recipient reads a message.

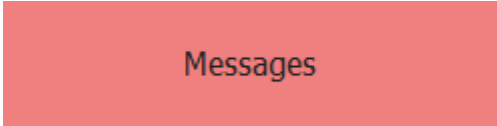
The function of sending messages is only available in tile view.

Call

You can open the application in the basic display in tile view of the AIP2. Use the button *Messages* at the bottom left to open the application.



If there are unread messages, the button *Messages* is highlighted in a different color.



You can refresh the display. See paragraph [Configuration](#).

Dialog *Message overview*

The dialog *Message overview* shows an overview of the messages the terminal is involved in. In general, the function shows all active message histories i.e. that are not closed. In addition all messages (histories) are shown that were closed within the last five days (default setting – see also paragraph [Configuration](#)). Closed messages are grayed out in the Message overview.

The table is sorted as follows:

- First all active messages are shown. The active messages are sorted in descending chronological order based on the time of the last update.
- Then the closed messages are shown. These messages are also sorted in descending chronological order based on the time of the last update. Do not confuse the last update with the time the message was closed.

The columns available in the overview are described below.

No title

The column *No title* shows the status of the message graphically. The table below describes the possible symbols in the column *No title*.

Symbol	Meaning
	If the first message of a message history addressed to the terminal is not marked read, the symbol is highlighted in red .
	If a message of the message history is not marked read by the terminal, the symbol is highlighted in red .
	The terminal has sent a <i>new</i> message to a recipient (person, function group). The new message is not yet marked read.
	The first message of the message history is marked read.  This symbol is even shown, if the dialog Message history includes messages that are not marked read.
	The message history is closed. The message is additionally grayed out.

Date

The column *Date* shows the date of the most recent message of the message history.

Time

The column *Time* shows the time of the most recent message of the message history.

Communication partner

The column *Communication partner* shows the communication partner of the most recent message of the message history.



If a message has been forwarded, the recipient of the forwarded message is displayed as communication partner.

Subject

The column *Subject* shows the subject of the message that was entered on creating a new message.

Text

The column *Text* shows an extract of the complete message text.

Once the message history is closed, the column *Text* shows the final comment.

Buttons in the dialog box *Message overview*

Use the button  to refresh the message overview.



Show message history

Use the button *Message history* to show the message history of a selected message.



New message

Use the button *New message* to create a new message.

Dialog - Message history

The upper part of the dialog box *Message history* shows all messages of the message history. The messages are displayed in a descending chronological order. The lower part of the dialog box shows the complete message content of a message selected in the upper part. By default the message on top of the message history is selected (most recent message).

By default, incoming messages are shown in black font color, outgoing messages are shown in gray font color.

The columns available in the upper part of the dialog box are described below.

No title

The first column *No title* indicates, if it is an incoming or an outgoing message. The possible symbols are described below.

Symbol	Meaning
	It is an incoming message.
	It is an outgoing message.
	It is a message forwarded to another communication partner.

No title

The second column *No title* shows the status of the message graphically. The possible symbols are described below.

Symbol	Meaning
	The message is marked read.
	The message is not marked read.

Date

The column *Date* shows the date of the message.

Time

The column *Time* shows the time of the message.

Communication partner

The column *Communication partner* shows the communication partner.

Text

The column *Text* shows the text of a message.

Buttons in the dialog box *Message history*



Read

You use the button *Read* to mark a message read.

The button is only active with a message addressed to the terminal not yet being marked read.

Clicking on this button does not open a dialog. The message is directly marked read.



Answer

Use the button *Answer* to answer a message.

The button is only active with a message addressed to the terminal being marked.



Forward

Use the button *Forward* to forward a message.

The button is only active with a message addressed to the terminal being marked.



Close message history

Use the button *Close message history* to close a message history.

The button is only active, if the message history has not yet been closed.



Each person involved in the message history can close the message history. This is not restricted to the person who has created a new message.

If you close a message history containing unread messages , these messages are automatically marked read . At the same time, the time stamp of the message history is updated and the closed message history is displayed on top of the list.

If all messages of a message history are already marked read on closing the message history, the message is only set to the status closed, but the time stamp of the message history is not updated. The closed message therefore stays at its position in the list.

Reporting dialogs

Overview of the fields you can edit in each reporting dialog

	New message	Answer	Forward	Close
Type of recipient	✓	read-only	✓	not available
Recipient	✓	read-only	✓	not available
Subject	✓	read-only	read-only	not available
Message/ Final comment	✓	✓	✓	✓
Mail delivery	✓	✓	✓	not available

Field descriptions

Type of recipient

In the field *Type of recipient* you can select the type of recipient. The following types of recipients are available:

- Function group
- Person



A terminal cannot be a recipient.

Recipient (*Type of recipient*: Function group)

In the field *Recipient* you can enter the name of the function group.



The search function of the field *Recipient* does not filter. The system shows all created function groups. The system does not verify, if the recipient can display the message. Persons without sufficient authorizations might be recipients of a message without being able to display the message.

Recipient (*Type of recipient: Person*)

In the field *Recipient* you enter the unique personnel number of the recipient.



The search function of the field *Recipient* does not filter. The system shows all created persons. The system does not verify, if the recipient can display the message. Persons without sufficient authorizations might be recipients of a message without being able to display the message.

Subject

In the field *Subject* you enter the subject of a message. You can use a maximum number of 50 characters.

Message (only *New message*)

In the field *Message* you enter the message content. You can use a maximum number of 600 characters.

Mail delivery

If you select this function, an e-mail is sent in addition to the message.



You can only use the option *Mail delivery* in connection with the types of recipient *Person* and *Function group*. The necessary configurations for sending e-mails require the escalation management license. The documentation of the escalation management provides further information on the configurations.

Message (in the dialog box *Forward*)

If you forward a message, you can use the field *Message*. You can enter a comment on the forwarded message in the field *Message*. The original message content is not changed.

Final comment

If you close a message history, you can use the field *Final comment*. You can enter a final comment in this field.

Configuration

Show closed message histories

In the configuration file *hytnrcfg.ini* you can set the following configuration:

```
[ HMS->OPTIONS 0 ]
```

```
; Show finished message histories of the last 5 days.
```

```
NUMBER-OF-DAYS-OVERVIEW=5
```

Display of new messages

On updating the terminal status, the server checks if there is a new message for the terminal. In this case, the button is highlighted in color. Define the update cycle of the terminal status in the MOC application *Terminal configuration* in the field *Cycle duration of status messages*.



"Notification" is done by cyclic polling. On configuring the *cycle duration of status messages*, you should therefore take into consideration the entire system and the shop floor clients. We do not recommend a configuration less than two minutes.

Closing the dialog box *Message overview* does not automatically update the button. The button can still be displayed in color (red), although all messages have been read. Only with the following status update, the color is set back.

Configuration file hms.ini



In the configuration file hms.ini you can define the display of the entries in the *Message overview* and in the message history. Changes to the configuration file hms.ini can lead to unpredictable and unwanted behavior.

4 Function Groups

Overview

HYDRA menu	Master data → Escalation management → Function groups
FEDRA menu	Detailed scheduling → Master data → Escalation history
Transaction code	escfg
Function authorization	escfg

Purpose

You use this function to create or modify function groups in the system.

Integration

Use function groups in the escalation management to

- notify several recipients at a time;
- identify specific recipients within a group (e.g. a group of maintenance engineers) that meet special requirements, e.g. the recipient is present.

Requirements

You have defined the recipients as person in the [HR master data](#) and:

- if you want a notification via e-mail, you have created the e-mail address of the person;
- If you want a notification in the MOC, you have linked the user in the [User administration](#) to a person of the HR master.

Field descriptions

Function

The function is the name of the function group. If function groups already exist, you can select the group via the detail selection.

Priority

The priority defines the notification order within a function group. You can assign the same priority to several persons within a function group.

The highest priority is "1", the lowest priority is "999". The priority helps the system to deliver the messages.

Personnel number

The personnel number identifies the person that should be notified.

Check console

If you have checked the option, the system will verify if the person is logged on to the client. If the person is not logged on, the system verifies the person with the next priority level. You use this setting, if you do not use the "time and attendance" system. In case of a negative result, i.e. the person is not logged on to the client/console, the message is not delivered in an alternative form (text message, e-mail...).

Example:

Three maintenance engineers work in three different shifts. If you assign within a group the same priority to all three of them, only the engineer actually logged in to the console receives the message.

If you want to use the function "Check console", you must have entered the personnel number in the User administration.

Check attendance

If you have checked the option "Check attendance", the system will verify, if the respective employee is logged on to the PZE system "time and attendance" (only in connection with PZE-BP).

The escalation management will suppose that the person is present, if the following PZE events occur: In, break, entrance before clocking in (with access control system ZKS).

There is an AND-link between the settings "Check console" and "Check attendance". If both options are checked, the person only receives messages when he or she is present AND logged in to the console.

That is the reason why we recommend to check the option "console" OR the option "attendance".



The system cannot send a message to a function group, if no one of this group is logged in to the console or is logged in to the PZE system "time and attendance".

In this case the escalation management will generate the event "NO_PERSON_AVAILABLE".

You should assign a dispatcher group to this event or you should search for this event on a daily basis in order to find cases left aside.